Iyad Abid

Lausanne, Switzerland | abid15iyad@gmail.com | iyad-coder.github.io | linkedin.com/in/iyad-abid-387715368 | github.com/Iyad-coder

Education

École Polytechnique Fédérale de Lausanne (EPFL) — Lausanne, Switzerland
B.Sc. in Microengineering (robotics & embedded systems) — *Expected Jul 2027*

- **Coursework:** Microcontrollers, Electronics II, Numerical Analysis, Complex Analysis & PDEs, Electrical Engineering I, Materials

Hackathons & Awards

3rd Place, HackTurin (hardware track) | *Drone payload, RGB + thermal imaging* | GitHub — Oct 2026

- Built a drone-mounted RGB + thermal camera payload that detects solar-panel defects during flight

2nd Place, LauzHack × Huawei | *Mixture-of-Experts LLM serving* — May 2026

- Designed a dynamic MoE expert-placement algorithm (dual EMA load estimation, replication hedging, bandwidth-cost gating) that beat DeepSeek's baseline by **14.2%**

RoboHack 2026, EPFL | *SO-101, Hugging Face LeRobot, PyTorch* | GitHub — May 2026

- Recorded teleoperation demonstrations, published the dataset on Hugging Face and trained an ACT imitation-learning policy for autonomous leaf picking

Accepted, AI Hackathon co-organized by EPFL AI Team, Switzerland — Nov 2026

Experience & Leadership

Electronics Lead, Drone Team — Oct 2026 – Present
EPFL Xplore (European Rover Challenge) — *Lausanne, Switzerland*

- Lead the electronics of the team's competition drone: dual 6S batteries with an electrical hot-swap system, Pixhawk 6X flight controller, Jetson Orin Nano companion computer and 4 LiDARs

Engineering Consultant — Oct 2026 – Present
Junior Entreprise EPFL — *Lausanne, Switzerland*

- Conduct technical feasibility and patentability studies for client hardware projects (under NDA)
- Evaluate component and sensor choices, run prior-art searches and deliver cost estimates and recommendations

Math Tutor
CVAJ — *Lausanne, Switzerland*

- Tutor high-school students in mathematics

Projects

Autonomous Rover, Navigation & Sensor Fusion | *ESP32, ROS 2, Gazebo, OpenCV, EKF*

- Designed the custom PCB and full electronics for an ESP32 differential-drive rover (4 DC motors, 5 servos, IMU, LiDAR, ultrasonic)
- Fused IMU, LiDAR and ultrasonic data with an Extended Kalman Filter; calibrated the IMU and fixed odometry covariance so the filter accepted wheel odometry
- Built HSV target detection and a LiDAR gap-finding state machine (search, align, approach, avoid, arrive), validated in Gazebo before deployment

Edge Thermal-Vision Drone Payload | *Arduino UNO Q, MLX90621, Python* — Oct 2026 – Present

- Building a thermal-imaging payload on an Arduino UNO Q (Linux MPU + MCU) with a 16×4 IR array, rendering live heat maps on-device and fusing them with RGB frames

Humanoid Robot | *Software* — In progress

- Building a humanoid robot and its software stack, with an on-board perception pipeline as the next milestone

Stitchmeter, Surgical Suturing Assist | *Compliant mechanisms*

- Designed a fully mechanical suturing-assist device using flexure guides to help surgeons place consistent stitches

Skills

Programming: Python, C/C++
Embedded & Hardware: ESP32, Arduino (UNO Q), Pixhawk, Jetson, I^2C, PWM, motor drivers, IMU calibration, PCB design
AI & Robotics: PyTorch, Hugging Face LeRobot, imitation learning (ACT), Mixture-of-Experts, ROS 2, Gazebo, OpenCV, Kalman filtering
Tools: Linux (Ubuntu), Git
Languages: French (native), Arabic (native), English, Spanish